Final Review

ECE 447: Data Analysis and Machine Learning for Engineers

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Final Materials

Lectures

- Linear Models
- Probabilistic ML
- Regularized Optimization
- KNN and Clustering
- Unsupervised Learning

You should know the key concepts from **Seminars** to be able to understand the **Lectures**.

You should know the key concepts (understanding of how models perform) from **Notebooks**.

Probability Mass Functions

A *discrete* random variable is fully characterized by its *probability mass function (pmf)*.

$$\mathscr{T} \text{ is discrete.}$$

For $x \in \mathscr{T}$*, the pdf is*

$$P(\boldsymbol{X} = x) \geq 0$$

and

$$\sum_{x \in \mathscr{T}} P(\boldsymbol{X} = x) = 1$$

Probabilities

Joint probability $p(x, y)$

Marginal probability $p(x)$

Conditional probability $p(x \mid y)$

How Can Joint, Marginal, and Conditional Probabilities Relate?

Product rule:

$$p(x, y) = p(y \mid x)p(x)$$

Bayes' rule:

$$\underbrace{p(x \mid y)}_{\textit{posterior}} = \frac{\overbrace{p(y \mid x)}^{\textit{likelihood}}\overbrace{p(x)}^{\textit{prior}}}{\underbrace{p(y)}_{\textit{evidence}}}$$

2D Gaussian Distributions

A Sample from random variable $\boldsymbol{X}$:

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

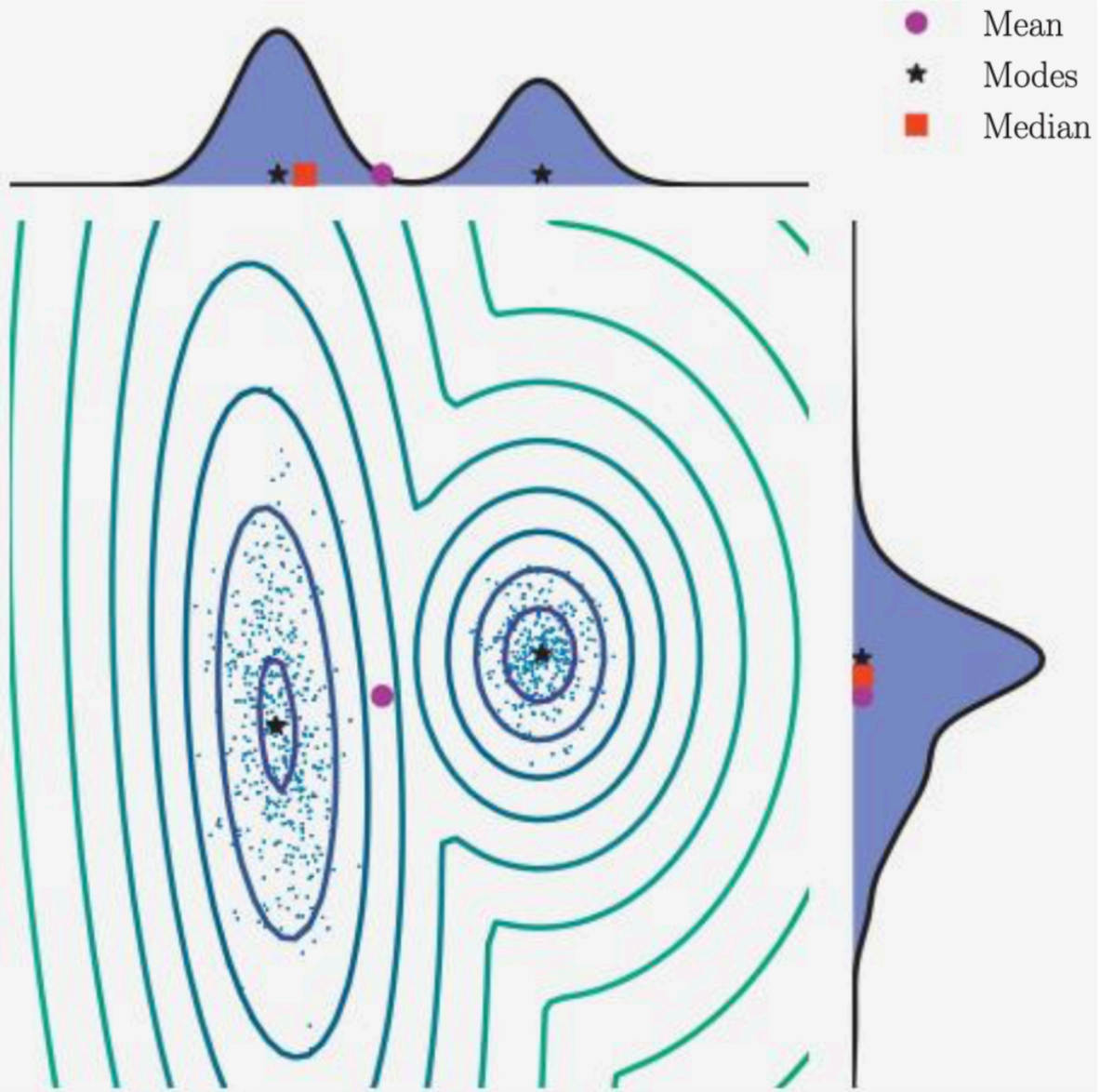
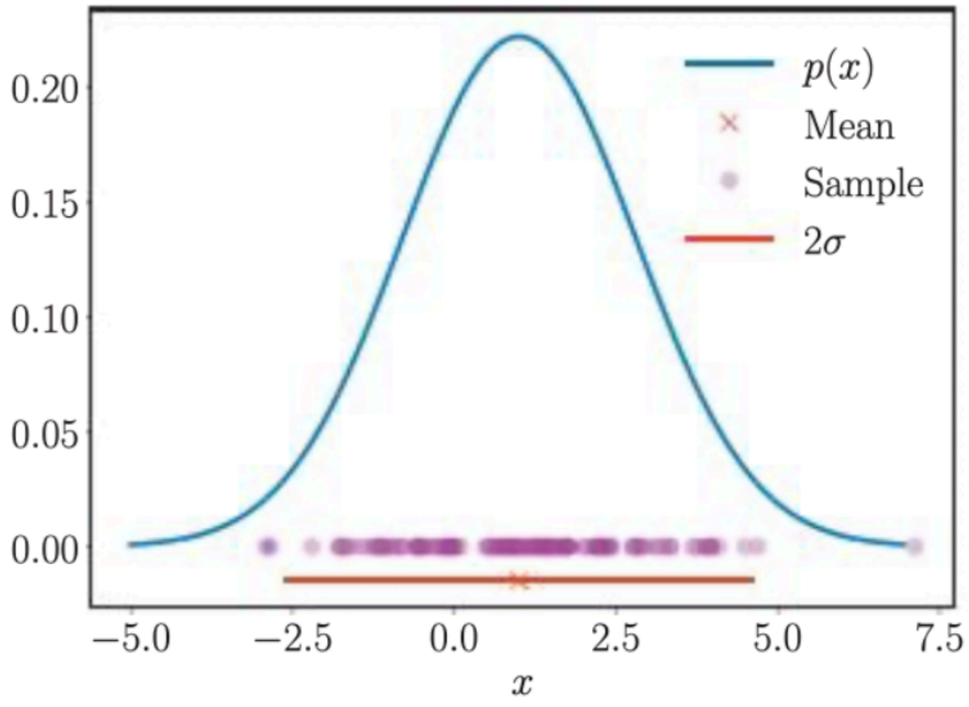


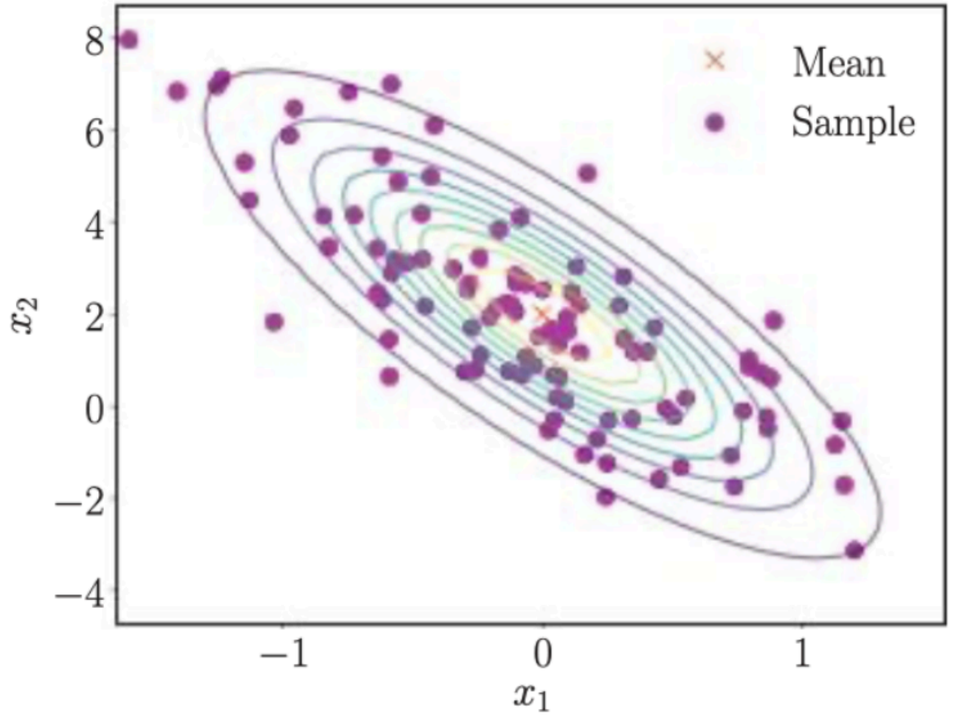
figure from Mathematics for Machine Learning textbook.

Gaussian Distribution Visualization

1D case



2D case



$$p(\boldsymbol{x}_1, \boldsymbol{x}_2) = \mathcal{N}\left(\begin{bmatrix} \boldsymbol{\mu}_{x_1} \\ \boldsymbol{\mu}_{x_2} \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Sigma}_{x_1x_2} & \Sigma_{x_1x_2} \\ \boldsymbol{\Sigma}_{x_2x_1} & \Sigma_{x_2x_2} \end{bmatrix}\right)$$

Sum of Gaussian Random *Variables*

If X, Y are independent Gaussian random variables with

$$p(\boldsymbol{x}) = \mathcal{N}(\boldsymbol{x} \mid \boldsymbol{\mu}_x, \boldsymbol{\Sigma}_x) \qquad p(\boldsymbol{y}) = \mathcal{N}(\boldsymbol{y} \mid \boldsymbol{\mu}_y, \boldsymbol{\Sigma}_y)$$

then

$$p(\boldsymbol{x} + \boldsymbol{y}) = \mathcal{N}(\boldsymbol{\mu}_x + \boldsymbol{\mu}_y, \boldsymbol{\Sigma}_x + \boldsymbol{\Sigma}_y)$$

Linear Transformations of Random Variables

Consider a Gaussian distributed random variable $X \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Let Y be a random variable such that $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}$. Then,

$$\mathbb{E}[\boldsymbol{y}] = \mathbb{E}[\boldsymbol{A}\boldsymbol{x}] = \boldsymbol{A}\mathbb{E}[\boldsymbol{x}] = \boldsymbol{A}\boldsymbol{\mu}$$

$$\mathbb{V}[\boldsymbol{y}] = \mathbb{V}[\boldsymbol{A}\boldsymbol{x}] = \boldsymbol{A}\mathbb{V}[\boldsymbol{x}]\boldsymbol{A}^\top = \boldsymbol{A}\boldsymbol{\Sigma}\boldsymbol{A}^\top$$

$$p(\boldsymbol{y}) = \mathcal{N}(\boldsymbol{y} \mid \boldsymbol{A}\boldsymbol{\mu}, \boldsymbol{A}\boldsymbol{\Sigma}\boldsymbol{A}^\top)$$

Linear Regression — Model

Training set

Given a set of data points $\{\boldsymbol{x}_i, y_i\}_{i=1}^n$,

$y_i \in \mathbb{R}$ output variable

$\boldsymbol{x}_i \in \mathbb{R}^p$ input vector (regressor)

The goal of the regression is to *estimate the parameter* $\boldsymbol{\theta}$ *from data modelled as*

$$y = f_{\boldsymbol{\theta}}(\boldsymbol{x}) = \boldsymbol{\theta}_1^\top \boldsymbol{x} + \theta_0 + \epsilon \qquad \epsilon \sim \mathcal{N}(0, \sigma^2)$$

$$y = \boldsymbol{\theta}^\top \boldsymbol{x} + \epsilon$$

Absorb θ_0 into the parameter

$$\boldsymbol{\theta} = \begin{bmatrix} \boldsymbol{\theta}_1 \\ \theta_0 \end{bmatrix}$$

$$\boldsymbol{x} \leftarrow [\boldsymbol{x}, 1]^\top$$

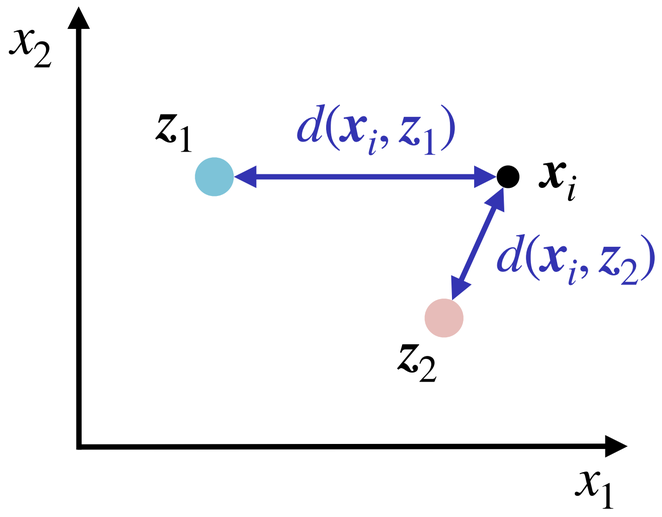
Linear Discriminant Function via Minimum Distance Classifier

Given a set of data $\{\boldsymbol{x}_i \in \mathbb{R}^p, y_i \in \{-1,1\}\}_{i=1}^n$, we define the classification decision as

$$\boldsymbol{x}_i \in c_k \text{ iff } d(\boldsymbol{x}_i, \boldsymbol{z}_k) < d(\boldsymbol{x}_i, \boldsymbol{z}_l) \qquad \text{for all } l \neq k$$

where $\boldsymbol{z}_k$ is a prototype of class c_k, and $d(\,\cdot\,,\cdot\,)$ is the distance function.

Intuition: Classify $\boldsymbol{x}_i$ to class c_1 if it is closer to the prototype of class c_1 than to c_2.



Optimization Objective for Minimum Distance Classifier

Another way to write the decision is via optimization, i.e., assign $\boldsymbol{x}$ to class c_k

$$k = \arg\min_j \; d(\boldsymbol{x}, \boldsymbol{z}_j)$$

Choose this as Minimum Euclidean Distance (MED)

$$f_k(\boldsymbol{x}) = \|\boldsymbol{x} - \boldsymbol{z}_k\|_2^2$$

The decision boundary would be:

$$f(\boldsymbol{x}) = f_k(\boldsymbol{x}) - f_l(\boldsymbol{x})$$

The classification criteria is:

$$\boldsymbol{x} \in c_k \text{ iff } f_k(\boldsymbol{x}) < f_l(\boldsymbol{x}) \; \forall l \neq k$$

Minimum Weighted Euclidean Distance

Euclidean Distance

$$d(\boldsymbol{x}, \boldsymbol{z}) = \left(\sum_{j=1}^{p} (x_j - z_j)^2 \right)^{\frac{1}{2}} = \|\boldsymbol{x} - \boldsymbol{z}\|_2$$

Weighted Euclidean Distance

$$d_{\boldsymbol{W}}(\boldsymbol{x}, \boldsymbol{z}) = \left(\sum_{j=1}^{p} w_j (x_j - z_j)^2 \right)^{\frac{1}{2}} = \|\boldsymbol{W}(\boldsymbol{x} - \boldsymbol{z})\|_2$$

$$= \left((\boldsymbol{x} - \boldsymbol{z})^\top \boldsymbol{W}^\top \boldsymbol{W} (\boldsymbol{x} - \boldsymbol{z}) \right)^{\frac{1}{2}}$$

$\boldsymbol{W}$: matrix with the weights w_j on its diagonal.

Independent and Identically Distributed (*i.i.d.*) Random Variables

The random variables $Y_1, Y_2, \ldots, Y_n$ are i.i.d. if they all have the same pdf/pmf.

Examples:

Gaussian distribution

$$Y_1, Y_2, \ldots, Y_n \text{ i.i.d. } \mathcal{N}(\mu, \sigma^2)$$

$$p_{Y_i}(y_i) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\frac{(y_i-\mu)^2}{\sigma^2}}$$

$$i = 1, \ldots, n$$

Bernoulli distribution

$$Y_1, Y_2, \ldots, Y_n \text{ i.i.d. } \mathrm{Bern}(p)$$

$$P(Y_i = 1) = p \qquad Y_i \in \{0,1\}$$

$$i = 1, \ldots, n$$

The likelihood Function

The likelihood function of n random variables $Y_1, \ldots, Y_n$, parameterized by $\boldsymbol{\theta} \in \mathbb{R}^p$, is a join pdf/pmf of the random variables written as a function of $\boldsymbol{\theta}$

$$L(\boldsymbol{\theta} \mid y_1, \ldots, y_n) = \quad p_{Y_1,\ldots,Y_n}(y_1, \ldots, y_n; \boldsymbol{\theta})$$

Ex. Let $Y_1, \ldots, Y_n$ be iid

$$\mathcal{N}(\mu, \sigma^2) \qquad \theta = (\mu, \sigma^2) \in \mathbb{R}^2$$

$$\mathcal{N}(\mu, 1) \qquad \theta = \mu \in \mathbb{R}$$

$$\mathrm{Bern}(p) \qquad \theta = p$$

The Maximum Likelihood (ML) Estimation (formal)

Given a likelihood function $L(\theta \,|\, y_1, \ldots, y_n)$, the ML estimator $\hat{\theta}_{\mathsf{ML}}$ of θ is a function of $y_1, \ldots, y_n$ that maximizes the likelihood function.

the joint probability distribution.

$$\hat{\theta}_{\mathsf{ML}} = \arg\max_{\theta} \; L(\theta \,|\, y_1, \ldots, y_n)$$

$$= \arg\max_{\theta} \; \log \; L(\theta \,|\, y_1, \ldots, y_n)$$

Lets Simplify the Log Likelihood

$$\log\ L(\mu \,|\, y_1, \ldots, y_n) = \log\left(\frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(\frac{-\sum_{i=1}^{n} (y_i - \mu)^2}{2\sigma^2}\right)\right)$$

$$= -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \mu)^2$$

$$\hat{\theta}_{\mathsf{ML}} = \arg\max_{\theta}\ \log\ L(\mu \,|\, y_1, \ldots, y_n) = \arg\max_{\theta}\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \mu)^2\right)$$

$$= \arg\max_{\theta}\left(-\sum_{i=1}^{n}(y_i - \mu)^2\right)$$

Have you seen loss this before?

$$= \arg\min_{\theta}\sum_{i=1}^{n}(y_i - \mu)^2$$

Gaussian Assumption Underlies Euclidean Loss (ℓ_2-norm)

When you ***minimize a Euclidean loss (ℓ_2-norm)****, you have placed a* ***Gaussian distributed assumption*** *on noise statistics.*

We got the following optimization objective

$$\hat{\theta}_{\mathsf{ML}} = \arg\min_{\theta} \sum_{i=1}^{n} (y_i - \mu)^2$$

for the model of data as $Y \sim \mathcal{N}(\mu, \sigma^2)$

$$y_i = \mu + \epsilon_i \qquad , \epsilon_i \sim \mathcal{N}(0,\sigma^2)$$

The noise of data is Gaussian distributed.

To find $\hat{\theta}_{\text{ML}}$, Take the Derivative Set to Zero

$$\frac{d}{d\mu} \log \; L(\mu \,|\, y_1, \ldots, y_n) = 0$$

$$\frac{d}{d\mu} \sum_{i=1}^{n} (y_i - \mu)^2 = 0$$

$$2 \sum_{i=1}^{n} (y_i - \mu) = 0$$

$$n\mu = \sum_{i=1}^{n} y_i \quad \Rightarrow \quad \boxed{\hat{\theta}_{\text{ML}} = \hat{\mu}_{\text{ML}} = \frac{1}{n} \sum_{i=1}^{n} y_i}$$

Maximum Likelihood estimate.

*Under **Gaussian** noise statistics, the estimate that **best explain your data** is sample mean.*

Probabilistic Perspective on Linear Regression (I)

Given a set of data points $\{\boldsymbol{x}_i, y_i\}_{i=1}^n$, suppose we construct the linear function

$$y_i = \boldsymbol{\theta}^\top \boldsymbol{x}_i + \epsilon_i \qquad , \boldsymbol{x}_i \in \mathbb{R}^p \qquad i = 1,\ldots,n$$

$$, \epsilon_i \sim \mathcal{N}(0,\sigma^2) \qquad \textit{iid}$$

Let $\boldsymbol{X} = \begin{bmatrix} -\boldsymbol{x}_1^\top - \\ -\boldsymbol{x}_2^\top - \\ \vdots \\ -\boldsymbol{x}_n^\top - \end{bmatrix} \in \mathbb{R}^{n\times p} \qquad \boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix} \in \mathbb{R}^n,$

We get the familiar linear equation $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\theta} + \boldsymbol{\epsilon}$.

Probabilistic Perspective on Linear Regression (II)

$$\boldsymbol{y} = \boldsymbol{X\theta} + \boldsymbol{\epsilon}$$

Goal: Find the maximum likelihood of your parameter $\hat{\boldsymbol{\theta}}_{\text{ML}}$.

What is the probability distribution of $\boldsymbol{y}$?

$$\boldsymbol{y} \sim \mathcal{N}(\boldsymbol{X\theta}, \sigma^2 \boldsymbol{I}) \qquad p_{\boldsymbol{Y}}(\boldsymbol{y}) = \frac{1}{(2\pi\sigma^2)^{n/2}} e^{\frac{-1}{2\sigma^2}\|\boldsymbol{y}-\boldsymbol{X\theta}\|_2^2}$$

Log likelihood: $\log L(\boldsymbol{\theta}\,|\,\boldsymbol{y}) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\|\boldsymbol{y} - \boldsymbol{X\theta}\|_2^2$

$$\hat{\boldsymbol{\theta}}_{\text{ML}} = \arg\max_{\boldsymbol{\theta}} \log L(\boldsymbol{\theta}\,|\,\boldsymbol{y}) = \arg\max_{\boldsymbol{\theta}} -\frac{1}{2\sigma^2}\|\boldsymbol{y} - \boldsymbol{X\theta}\|_2^2$$

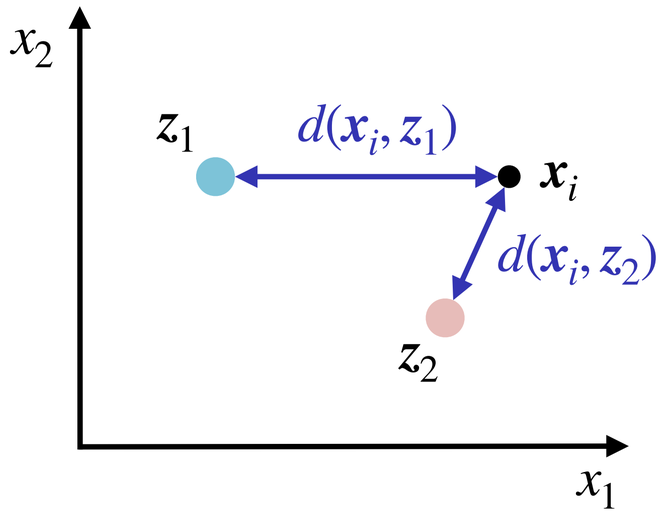
Maximum Likelihood (ML)!

$$= \arg\min_{\boldsymbol{\theta}} \frac{1}{2\sigma^2}\|\boldsymbol{y} - \boldsymbol{X\theta}\|_2^2$$

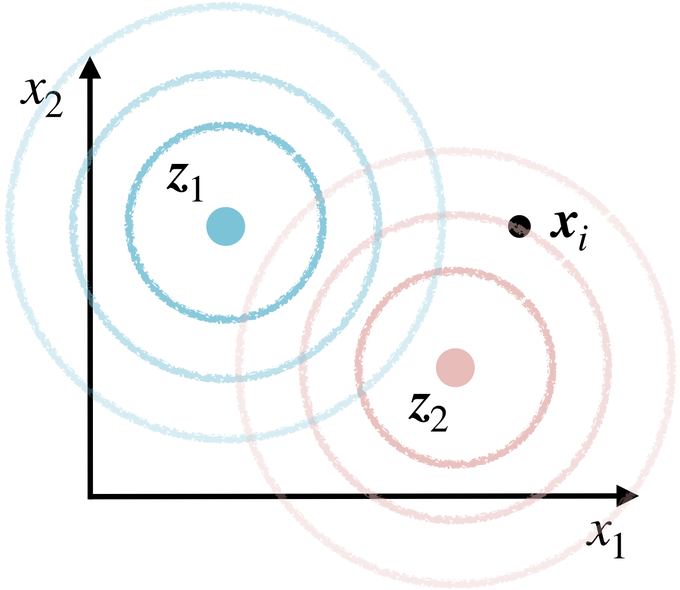
Least Square (LS)!

Two Perspectives on the Same Classification Method

Define a **distance metric** and assign to the class with **smallest distance** to.



Define a **probability distribution** and assign to the class with **maximum likelihood**.



Why Is This Perspective Useful?

You can derive lots of ML classifiers with only *one logic:*

*assign $\boldsymbol{x}_i$ to the **most probable class** to come from.*

Methods differ from one another in their **assumptions** to the **data distribution**.

Re-visit Minimum Euclidean Distance (MED) Classifier

Given a set of data $\{\boldsymbol{x}_i \in \mathbb{R}^p, y_i \in \{-1,1\}\}_{i=1}^n$, we define the classification decision as

$$\boldsymbol{x}_i \in c_k \text{ iff } p_k(\boldsymbol{x}) > p_l(\boldsymbol{x}) \qquad \text{for all } l \neq k$$

where $p_k(\boldsymbol{x}) = \mathcal{N}(\boldsymbol{z}_k, \sigma^2 \boldsymbol{I})$.

$$\frac{1}{(2\pi\sigma^2)^{p/2}} e^{\frac{-1}{2\sigma^2}\|\boldsymbol{x}_i - \boldsymbol{z}_k\|_2^2} > \frac{1}{(2\pi\sigma^2)^{p/2}} e^{\frac{-1}{2\sigma^2}\|\boldsymbol{x}_i - \boldsymbol{z}_l\|_2^2}$$

(take the log) $$\frac{-1}{2\sigma^2}\|\boldsymbol{x}_i - \boldsymbol{z}_k\|_2^2 > \frac{-1}{2\sigma^2}\|\boldsymbol{x}_i - \boldsymbol{z}_l\|_2^2 \quad \Rightarrow$$

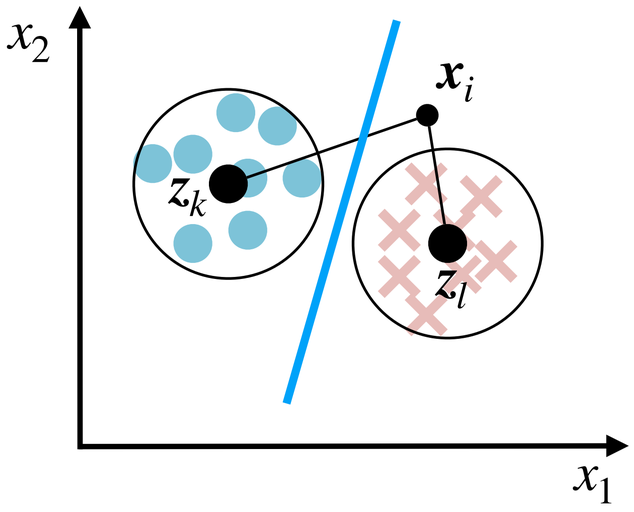
$$\boldsymbol{x}_i \in c_k \text{ iff}$$

$$\|\boldsymbol{x}_i - \boldsymbol{z}_k\|_2^2 < |\boldsymbol{x}_i - \boldsymbol{z}_l\|_2^2$$

This is **MED**. You have assumed classes are distributed as $\mathcal{N}(\boldsymbol{z}_k, \sigma^2 \boldsymbol{I})$.

Underlying Assumptions behind MED and WED

Minimum Euclidean Distance

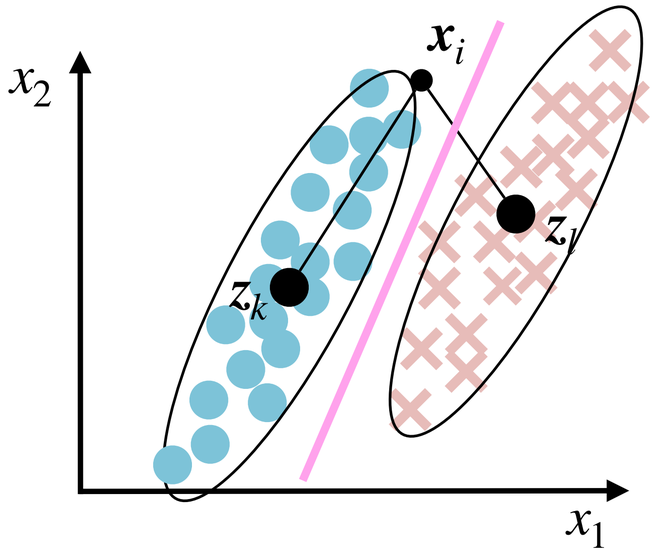


Features are not much correlated

The classes/features have similar unit/scale/variance

$$p_k(\boldsymbol{x}) = \mathcal{N}(z_k, \sigma^2 \boldsymbol{I})$$

Minimum Weighted Euclidean Distance



Features are correlated

The features have different unit/scale/variance

$$p_k(\boldsymbol{x}) = \mathcal{N}(z_k, \boldsymbol{\Sigma})$$

Bayesian 2-Class Classification

Lets remind ourselves on WED decision criteria on the board.

Let $\{y_i \in \{0,1\}, \boldsymbol{x}_i \in \mathbb{R}^p\}_{i=1}{}^n$. Assume

$$p(\boldsymbol{x} \,|\, y = c) = \mathcal{N}(\bar{\boldsymbol{x}}^c, \boldsymbol{\Sigma})$$

Goal: predict $p(y_i \,|\, \boldsymbol{x}_i)$.

Bayes' Rule:

$$\underbrace{p(y = c \mid \boldsymbol{x})}_{\textit{posterior}} = \frac{\overbrace{p(\boldsymbol{x} \,|\, y = c)}^{\textit{likelihood}}\overbrace{p(y = c)}^{\textit{prior}}}{p(\boldsymbol{x})}$$

Assign examples to the class with the **Maximum a Posteriori (MAP)** rule.

$$\boldsymbol{x}_i \in k \text{ iff } p(y = k \mid \boldsymbol{x}) > p(y = l \,|\, \boldsymbol{x})$$

for all $l \neq k$

MAP Decision Boundary

For prediction $\hat{c} = 1$.

$$p(y = 1 \mid \boldsymbol{x}) \geq p(y = 0 \mid \boldsymbol{x})$$

$$\frac{p(y = 1 \mid \boldsymbol{x})}{p(y = 0 \mid \boldsymbol{x})} \geq 1$$

$$\frac{p(\boldsymbol{x} \mid y = 1)p(y = 1)}{p(\boldsymbol{x} \mid y = 0)p(y = 0)} \geq 1$$

(take the log)

$$\underbrace{\log p(\boldsymbol{x} \mid y = 1) - \log p(\boldsymbol{x} \mid y = 0)}_{\text{This is Likelihood from WED.}} + \underbrace{\log \frac{\pi_1}{\pi_0}}_{\text{This is from prior.}} \geq 0$$

Regularization in Optimization

Consider the optimization problem for linear regression

$$\min_{\boldsymbol{\theta}} \quad \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\theta}\|_2^2$$

One can *regularize* the solution by adding an additional term

$$\min_{\boldsymbol{\theta}} \quad \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\theta}\|_2^2 \; + \; \mathscr{R}(\boldsymbol{\theta})$$

Restrict the space of solutions.

High value of $\mathscr{R}(\boldsymbol{\theta})$ for undesired solution

Low value of $\mathscr{R}(\boldsymbol{\theta})$ for desired solution.

Ridge Regression — ℓ_2-norm Regularized Linear Regression

Given a set of data points $\{\boldsymbol{x}_i \in \mathbb{R}^p, y_i\}_{i=1}^n$, find the parameters $\boldsymbol{\theta}$ that minimizes:

$$\boldsymbol{\theta}_{\text{Ridge}} = \arg\min_{\boldsymbol{\theta}\in\mathbb{R}^p} \frac{1}{2}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\theta}\|_2^2 + \frac{\lambda}{2}\|\boldsymbol{\theta}\|_2^2 = (\boldsymbol{X}^\top\boldsymbol{X} + \lambda\boldsymbol{I})^{-1}\boldsymbol{X}^\top\boldsymbol{y}$$

where $\boldsymbol{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{R}^n, \quad \boldsymbol{X} = \begin{bmatrix} -\boldsymbol{x}_1^\top- \\ -\boldsymbol{x}_2^\top- \\ \vdots \\ -\boldsymbol{x}_n^\top- \end{bmatrix} \in \mathbb{R}^{n\times p}.$

Useful regularization when the LS solution is not unique (e.g., $n \ll p$).

Ridge Regression — Gaussian Prior on Parameter

Consider the linear model

$$y_i = \boldsymbol{\theta}^\top \boldsymbol{x}_i + \epsilon_i \qquad , \epsilon_i \overset{iid}{\sim} \mathcal{N}(0,\sigma^2)$$

$$p(\boldsymbol{\theta}) \sim \mathcal{N}(0,\sigma_\theta^2 \boldsymbol{I})$$ *Prior on θ*

Goal: find the parameter that maximizes the posterior $p(\boldsymbol{\theta}\,|\,\boldsymbol{y})$.

likelihood ***prior***

posterior

$$p(\boldsymbol{\theta}\,|\,\boldsymbol{y}) = \frac{p(\boldsymbol{y}\,|\,\boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\boldsymbol{y})}$$

We are interested in maximum a posteriori (MAP) estimation to combine both likelihood and prior.

Ridge Regression — MAP Estimation

$$p(\boldsymbol{\theta}\,|\,\boldsymbol{y}) = \frac{p(\boldsymbol{y}\,|\,\boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\boldsymbol{y})}$$

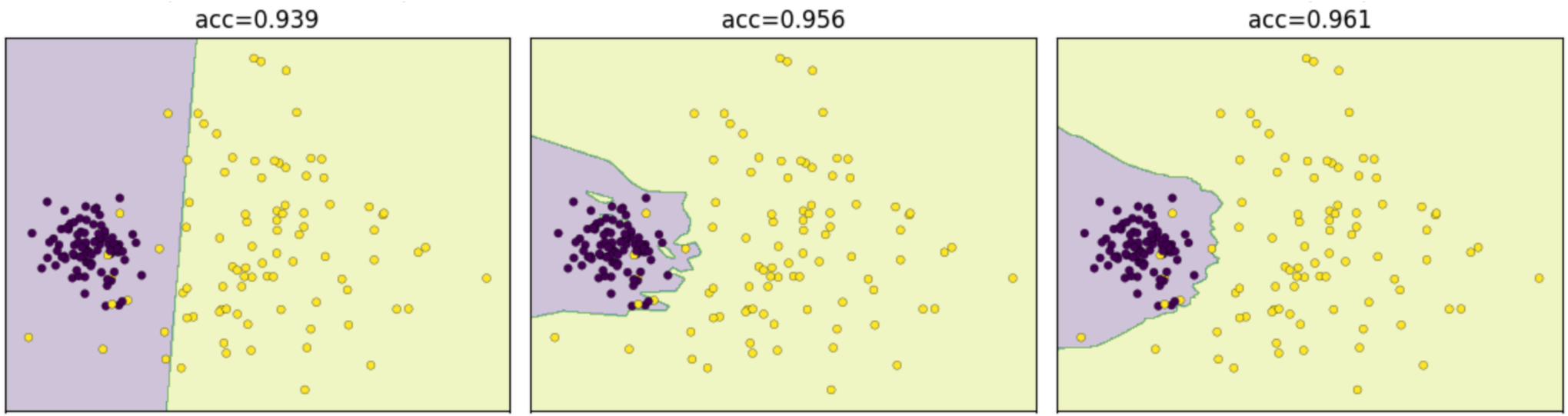
$$\begin{aligned}
\boldsymbol{\theta}_{\text{MAP}} &= \arg\max_{\boldsymbol{\theta}\in\mathbb{R}^p} \quad p(\boldsymbol{\theta}\,|\,\boldsymbol{y}) \\
&\quad \arg\max_{\boldsymbol{\theta}\in\mathbb{R}^p} \quad \log p(\boldsymbol{\theta}\,|\,\boldsymbol{y}) \\
&\quad \arg\max_{\boldsymbol{\theta}\in\mathbb{R}^p} \quad \log p(\boldsymbol{y}\,|\,\boldsymbol{\theta}) + \log p(\boldsymbol{\theta}) \\
&\quad \arg\min_{\boldsymbol{\theta}\in\mathbb{R}^p} \quad -\log p(\boldsymbol{y}\,|\,\boldsymbol{\theta}) - \log p(\boldsymbol{\theta}) \\
&\quad \arg\min_{\boldsymbol{\theta}\in\mathbb{R}^p} \quad \frac{1}{2\sigma^2}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\theta}\|_2^2 + \frac{1}{2\sigma_\theta^2}\|\boldsymbol{\theta}\|_2^2
\end{aligned}$$

When you regularize your parameter with ℓ_2-norm, you have indeed assumed a Gaussian prior on the parameter.

Multiple by σ^2 and denote $\lambda = \frac{\sigma^2}{\sigma_\theta^2}$, we get the same objective as regularized LS.

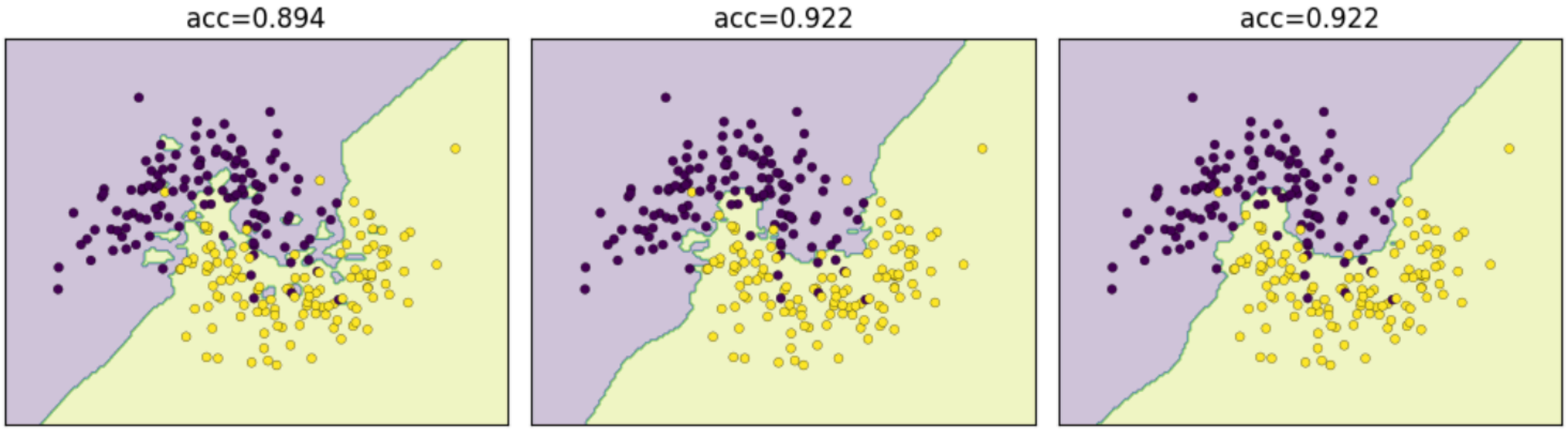
KNNs (I)

Which method is this? What is changing from left to right?



KNNs (II)

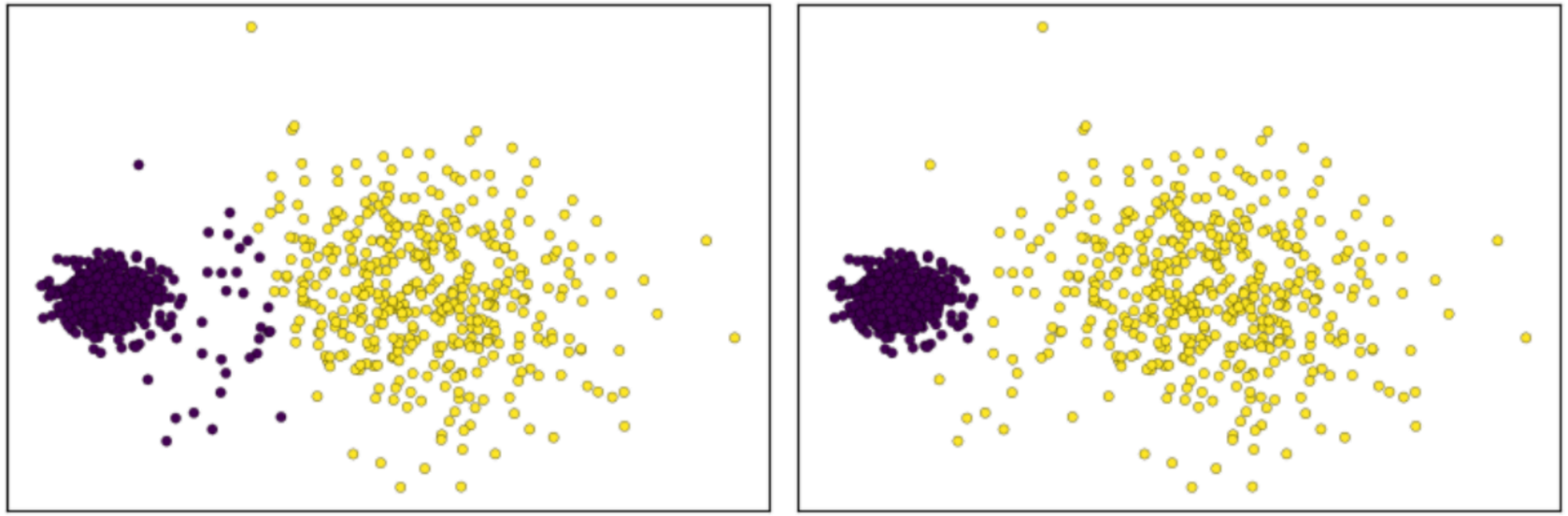
Which method is this? What is changing from left to right?



Clustering

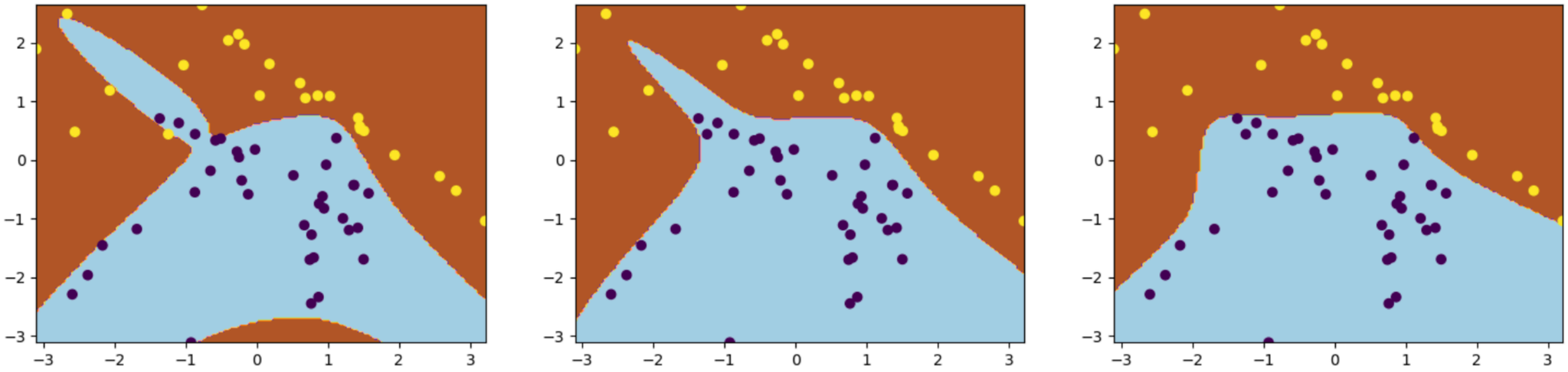
What is the difference between the two clustering methods?

Which one they are? (Kmeans, GMM, AgglomerativeClustering)



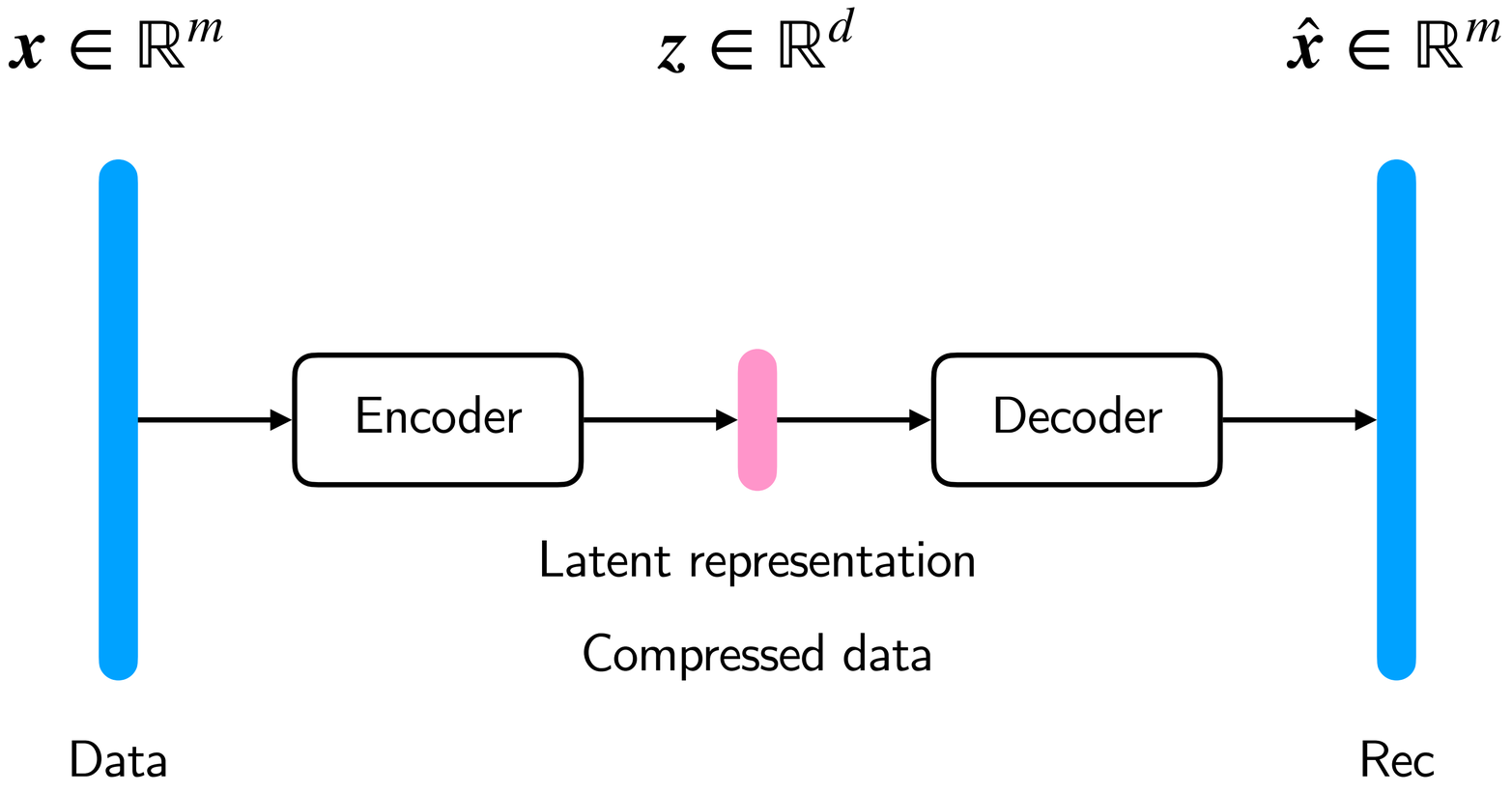
Logistic Regression with Polynomial Features

Which regularization is being used to which figure?



Dimensionality Reduction

Map data onto a low-dimensional space.



Linear Reconstruction Case — Setup (II)

Estimate the representation $\boldsymbol{z}^i$ for each $\boldsymbol{x}^i$.

$$\boldsymbol{X} \approx \boldsymbol{D}\boldsymbol{Z}$$

$$\begin{bmatrix} | & | & & | \\ \boldsymbol{x}^1 & \boldsymbol{x}^2 & \cdots & \boldsymbol{x}^n \\ | & | & & | \end{bmatrix}_{m\times n} \approx \begin{bmatrix} | & | & & | \\ \boldsymbol{d}_1 & \boldsymbol{d}_2 & \cdots & \boldsymbol{d}_p \\ | & | & & | \end{bmatrix}_{m\times p} \begin{bmatrix} | & | & & | \\ \boldsymbol{z}^1 & \boldsymbol{z}^2 & \cdots & \boldsymbol{z}^n \\ | & | & & | \end{bmatrix}_{p\times n}$$

Set of data
(each col is one example)

Set of features
(atoms, concepts, dictionary)

Set of coefficients (representations)
(Each col is for one example)

Linear Reconstruction Case — Optimization

Estimate the representation $\boldsymbol{z}^i$ for each $\boldsymbol{x}^i$.

$$\boldsymbol{X} \approx \boldsymbol{D}\boldsymbol{Z}$$

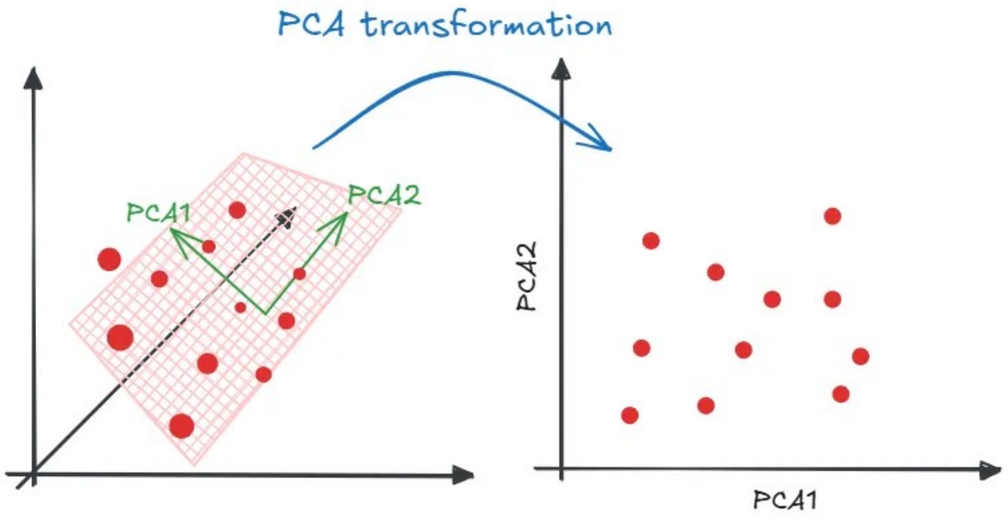
$$\min_{\boldsymbol{D},\, \boldsymbol{Z}} \quad \frac{1}{2}\|\boldsymbol{X} - \boldsymbol{D}\boldsymbol{Z}\|_2^2$$

Learn (estimate) a matrix $\boldsymbol{D}$ which can be used to map each data $\boldsymbol{x}^i$ to its representation $\boldsymbol{z}^i$.

*Linear dimensionality reduction methods differ in the **constraint** they impose on $\boldsymbol{D}$ or $\boldsymbol{z}$.*

Principal Component Analysis

Find a low-dimensional representation that explains for most variations in the data.



Data has **mean** of zero.

The representation is

$$z^i = D^\top x^i$$

The dictionary is orthonormal

$$D^\top D = I$$

$$\Rightarrow \quad \boxed{\begin{aligned} \min_{D,\, Z} \quad & \frac{1}{2}\|X - DZ\|_2^2 \\ \text{s.t.} \quad & D^\top D = I \end{aligned}}$$

Study well!

Good luck on your presentation and final exam!